# COMPSCI 3MI3 Assignment 2

## Academic integrity and assignment verification

Per the course outline, this assignment will be worth 3% of your final grade. You are encouraged to discuss these problems and potential solutions with your classmates, and you are welcome to ask about them during office hours. Your submitted solutions must be entirely your own work, perhaps informed by discussions you have had with your peers, the instructional team, or generative AI tools used in accordance with university policies and the course outline.

As part of the assessment process, students may be selected for a brief follow-up discussion regarding their submissions. These discussions are intended to verify authorship and understanding of the submitted work. Every student should expect to participate in at least one such discussion during the term.

Following a discussion, the instructional team may adjust an assignment grade based on the student's demonstrated understanding of the assessed material. Students who can clearly explain concepts that were not fully demonstrated in their written submission may receive additional credit. Conversely, if a student is unable to demonstrate an understanding consistent with the submitted work, this may result in a charge of academic dishonesty and the application of penalties as outlined in the McMaster Academic Integrity Policy.

The purpose of these discussions is both to support learning and to help ensure the academic integrity of course assessments.

## Submission instructions

You should submit your assignment solutions through Crowdmark no later than 11:59PM on Sunday, November 8th.

## Questions

All the questions in this assignment will refer to a language R whose syntax is given by the grammar below. You will be tasked with developing operational semantics for this language.

| | |
|---|---|
| `t ::=` | *Terms:* |
| `True` | *Constant True* |
| `False` | *Constant False* |
| `If t1 then t2` | *Conditional* |

*Questions 1 and 2 below are each worth 15 points.*
*Questions 3 and 4 are practice questions only. You will not score any points for these questions, and you can score full points for this assignment without submitting any solution to these questions.*

1. Define a one-step evaluation relation "→" for this language. The evaluation should be defined so that:
   - Whenever `t1` evaluates in multiple steps to `True`, and `t2` evaluates in multiple steps to `s`, the term `If t1 then t2` also evaluates in multiple steps to `s`
   - Whenever `t1` evaluates in multiple steps to `False`, any term of the form `If t1 then t2` evaluates in multiple steps to `True`

   Your solution should be a list of inference rules following the style of Pierce's language definitions. You should not include more inference rules than necessary. To score full points, your evaluation should satisfy the properties **a.** through **e.** listed below. You do not need to prove that these properties are satisfied.
   - **a.** Constant terms (`True, False`) cannot be evaluated
   - **b.** The one-step evaluation relation is deterministic (as a partial function on terms)
   - **c.** When evaluating the term "`If t1 then t2`", the guard `t1` should be evaluated before the conclusion `t2`
   - **d.** There are no stuck terms
   - **e.** Evaluation always terminates
2. Implement the language R as a data type RTerm in Haskell, with the one-step evaluation relation defined as evalOne :: RTerm -> Maybe RTerm
   (where evalOne returns Nothing for the terms R`True`, R`False`)
   Using the function evalOne, implement a big-step evaluation relation
   evalBig :: RTerm -> RTerm
   which returns R`True` for any term which eventually evaluates to R`True`, and R`False` for any other term.

3. For each of the statements (a) - (e) in Question 1, prove that your evaluation relation satisfies the relevant property.

4. Does your implementation of R in Haskell agree with the theoretical definition of the language? Can you prove it?